

Project Report:
Critical Thinking 3 – User Interface

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Project Report:

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This documentation is part of the Critical Thinking 3 Assignment from CSC372: Programming 2 at Colorado State University Global. This Project Report is an overview of the program's functionality and testing scenarios including console output screenshots. The program is coded in Java JDK-21; and is named Critical Thinking 3 (User Interface). The program is composed of only one class the `UserInterface` class that extends the `Application` class from the JavaFx library.

The Assignment Direction:

Option #1: **Creating a User Interface I**

Create a user interface that has a top bar that includes a menu. The menu should have four items.

- When the user selects the first menu option, then the date and time should be printed in a text box.
- When the user selects the second menu option, then the text box contents should be written to a text file named "log.txt."
- When the user selects the third menu item then the frame background color changes to random color hue of the color green. The menu option should display the initial random hue each time selected for a single execution of the program.

See https://www.w3schools.com/colors/colors_picker.asp Links to an external site.

- When the user selects the fourth menu option then the program exits.

Students must use appropriate version control for all programmatic assignments created. GIT repositories should be established and screen captures of repositories submitted with each assignment.

My notes:

- Instead of creating a menu top bar with the four items required, I created a menu top bar with a drop-down menu that contains the four required items.
- I added my own icon to the window frame – logo.png
- **For the source code please see `UserInterface.java`**

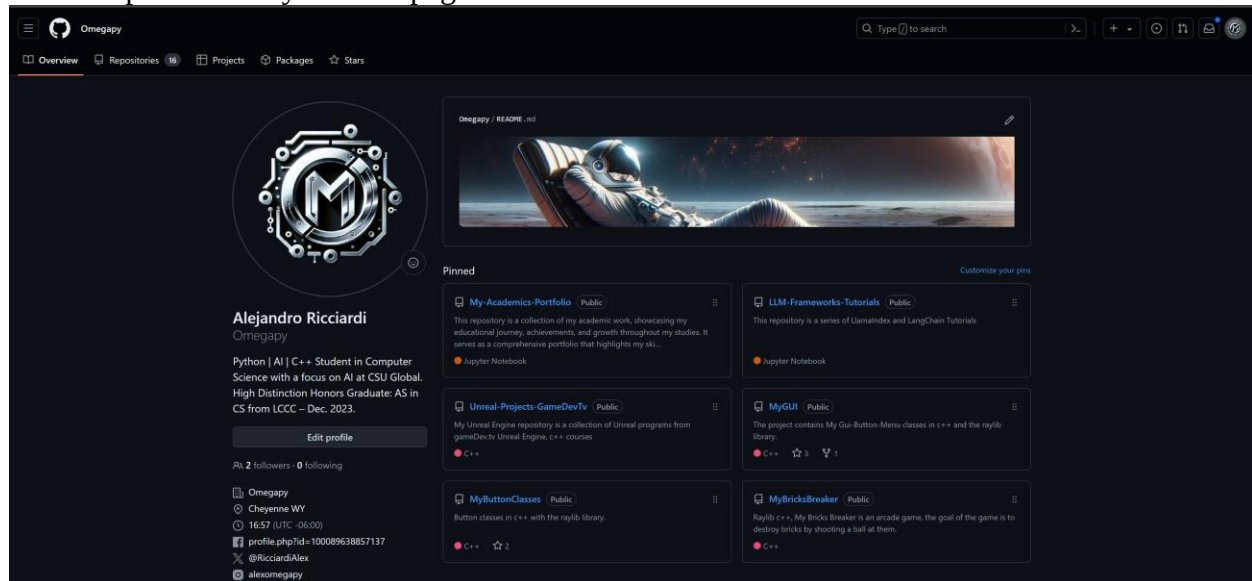
My Program Description:

The program is a small Java JavaFX application that provides a text area for user input and a menu bar with four options:

- Show Date and Time
- Saves the content of the text area to a file named "log.txt".
- Changes the background to a random shade of green.
- Exit

Git Repository

This is a picture of my GitHub page:



I use [GitHub](#) as my Distributed Version Control System (DVCS), the following is a link to my GitHub, [Omegapy](#).

My GitHub repository that is used to store this assignment is named [My-Academics-Portfolio](#) and the link to this specific assignment is:

<https://github.com/Omegapy/My-Academics-Portfolio/tree/main/Programming-2-CSC372/Critical-Thinking-3>

Classes Description:

- **The UserInterface Class:**

UserInterface class implements the main application window for a JavaFX application. It provides a text area for user input and a menu bar with various options. It also contains

Screenshots

Program Functionality and Testing Scenarios:

Figure 1
Welcome Window

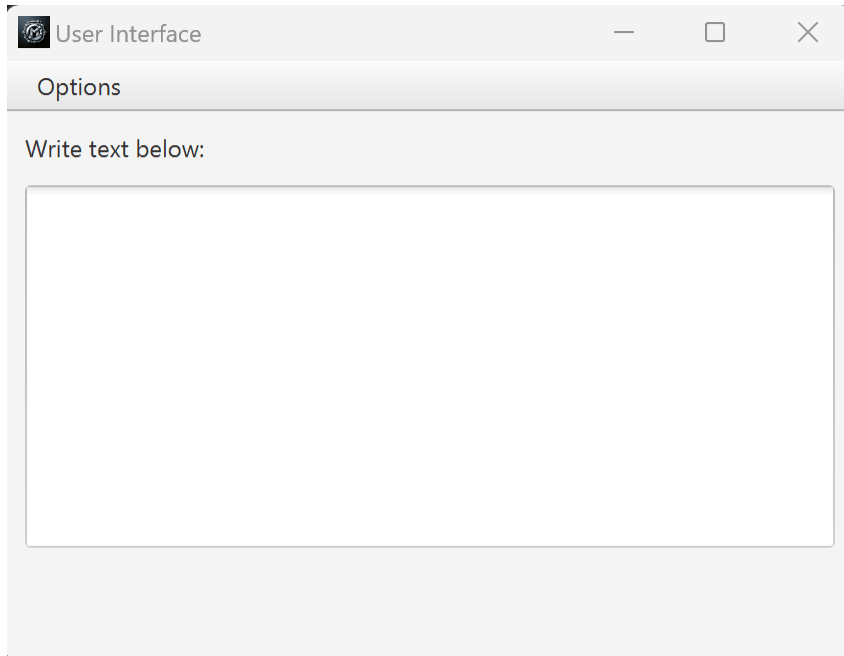
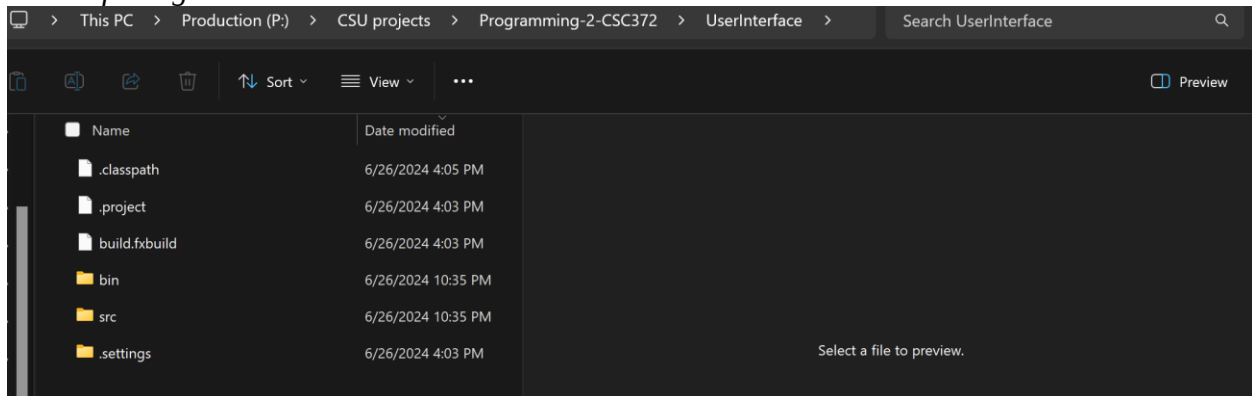


Figure 2
Folder for log.txt File



Note: No log.txt file in the folder.

Figure 3
User Entered Text

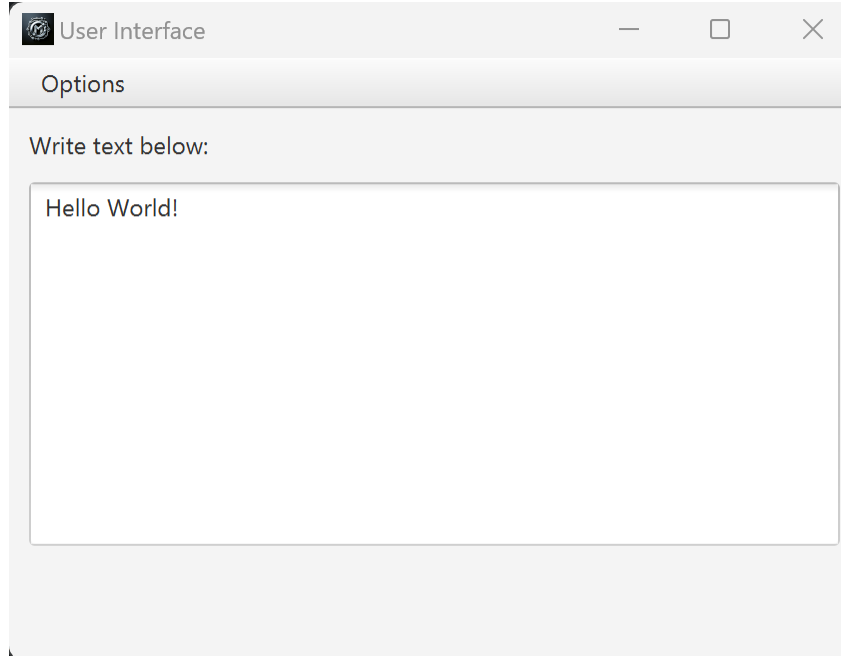


Figure 4
Drop Down Menu Option

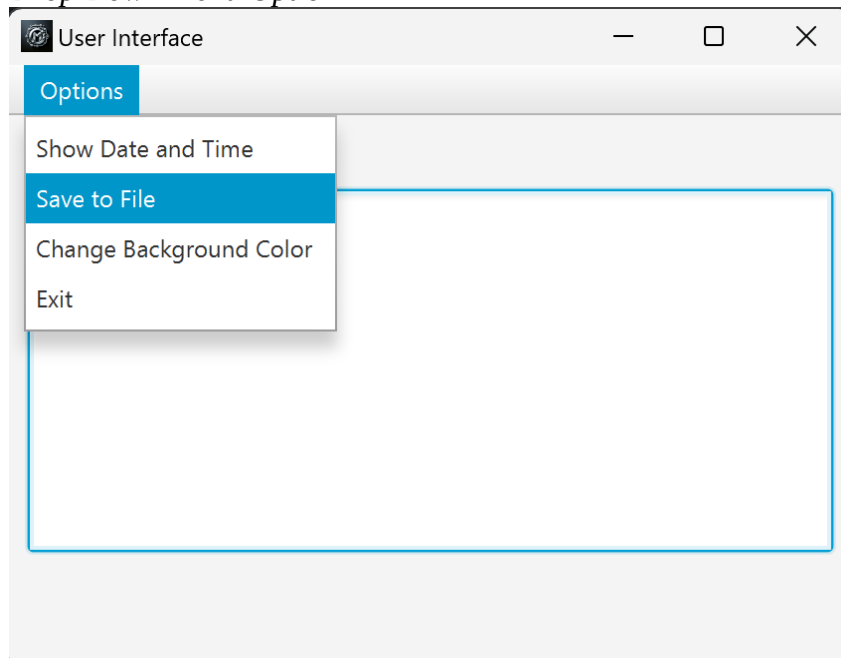
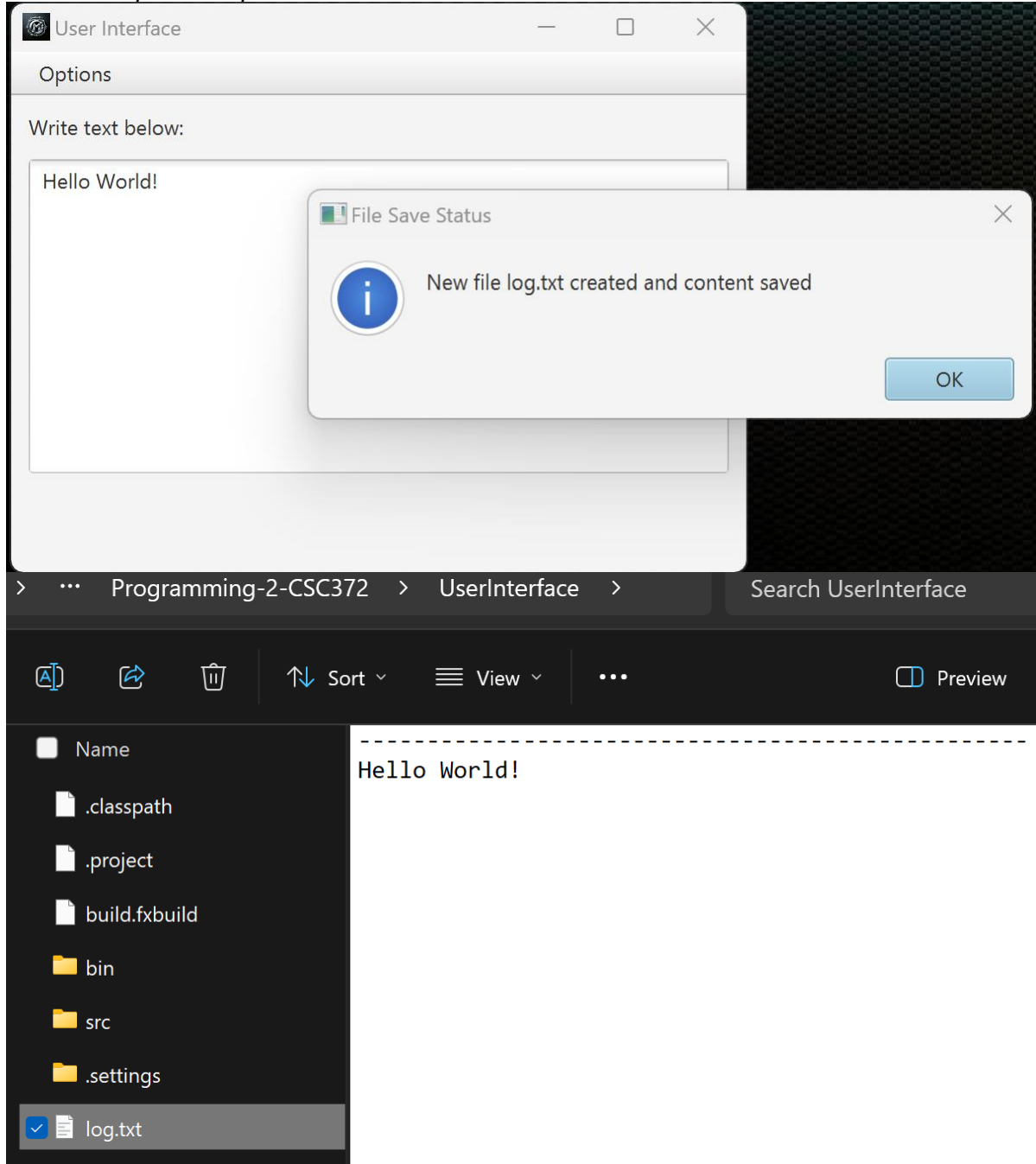


Figure 5
Save File Option Output



Note: After choosing the option “Save to File,” a log.txt file was created in the folder and “Hello World!” text was appended to the file.

Figure 6
Date and Time

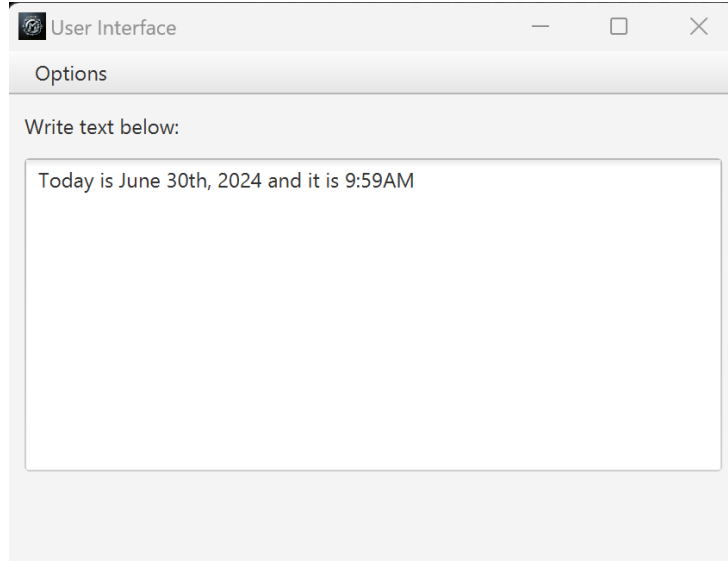
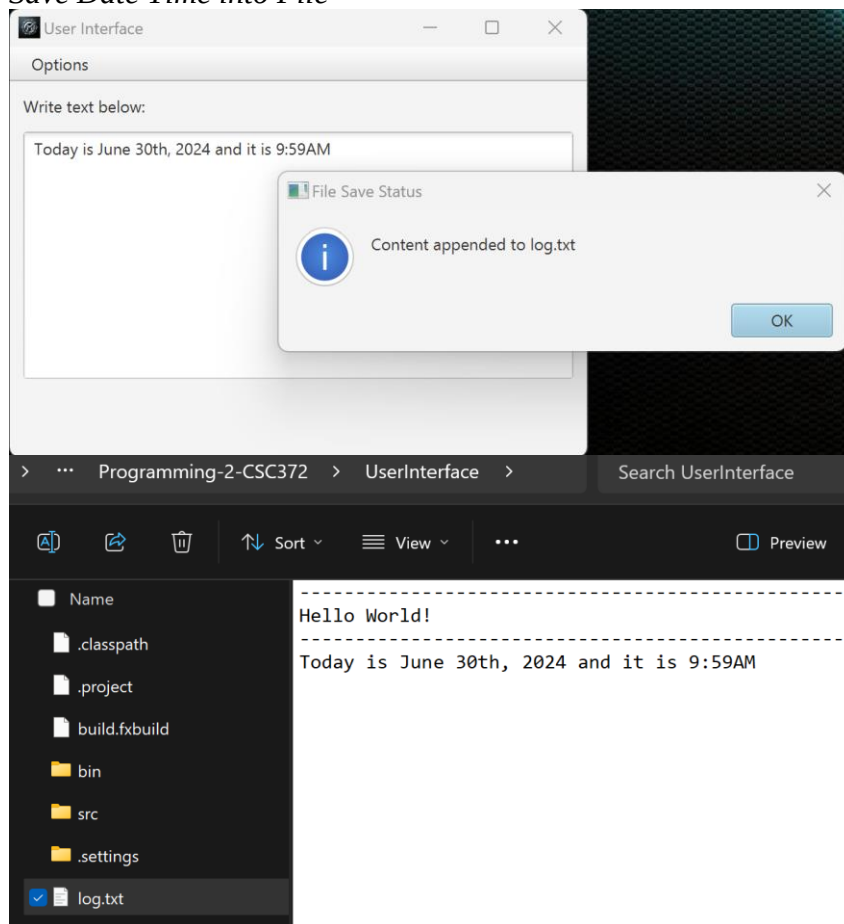
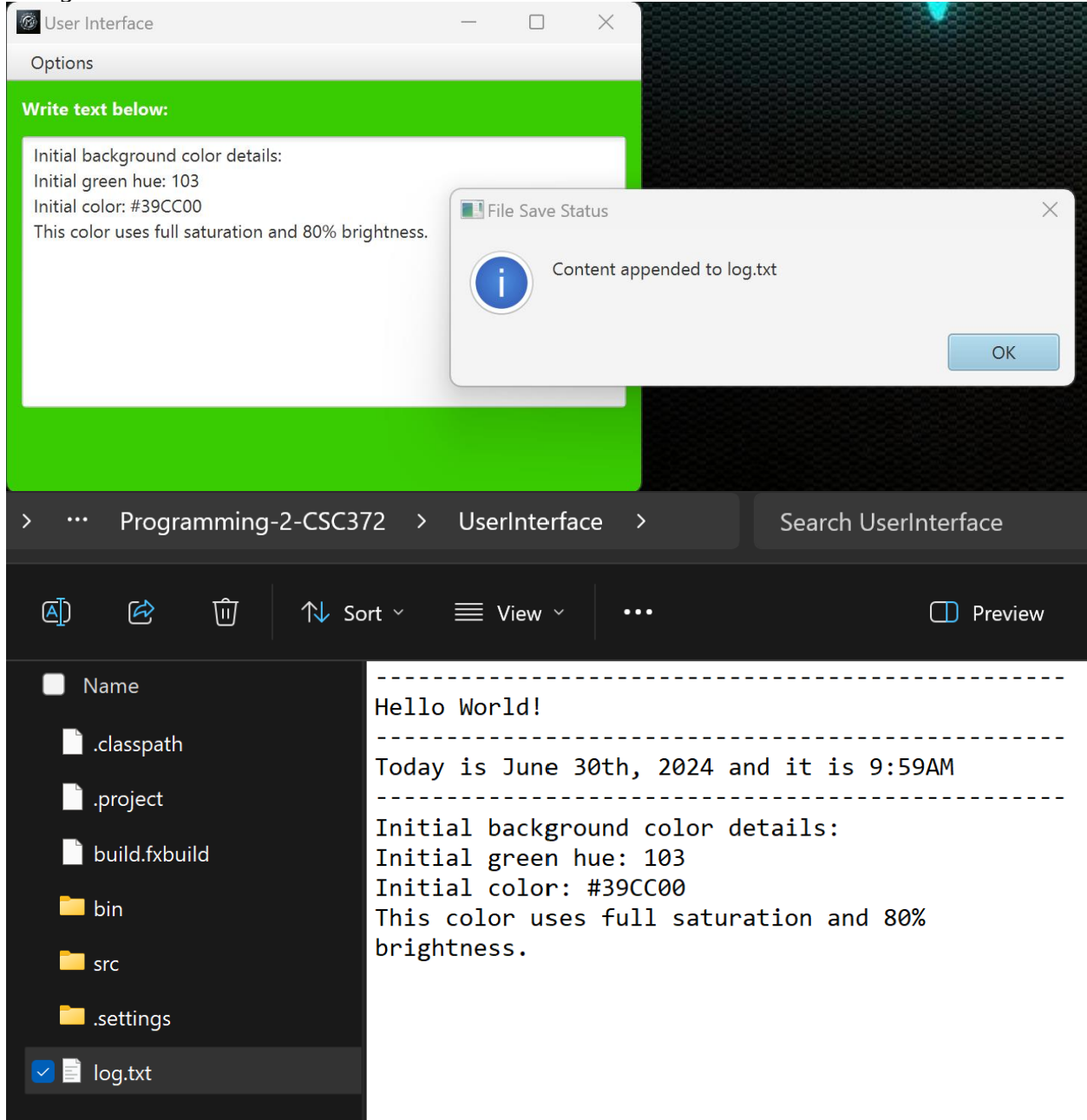


Figure 7
Save Date Time into File

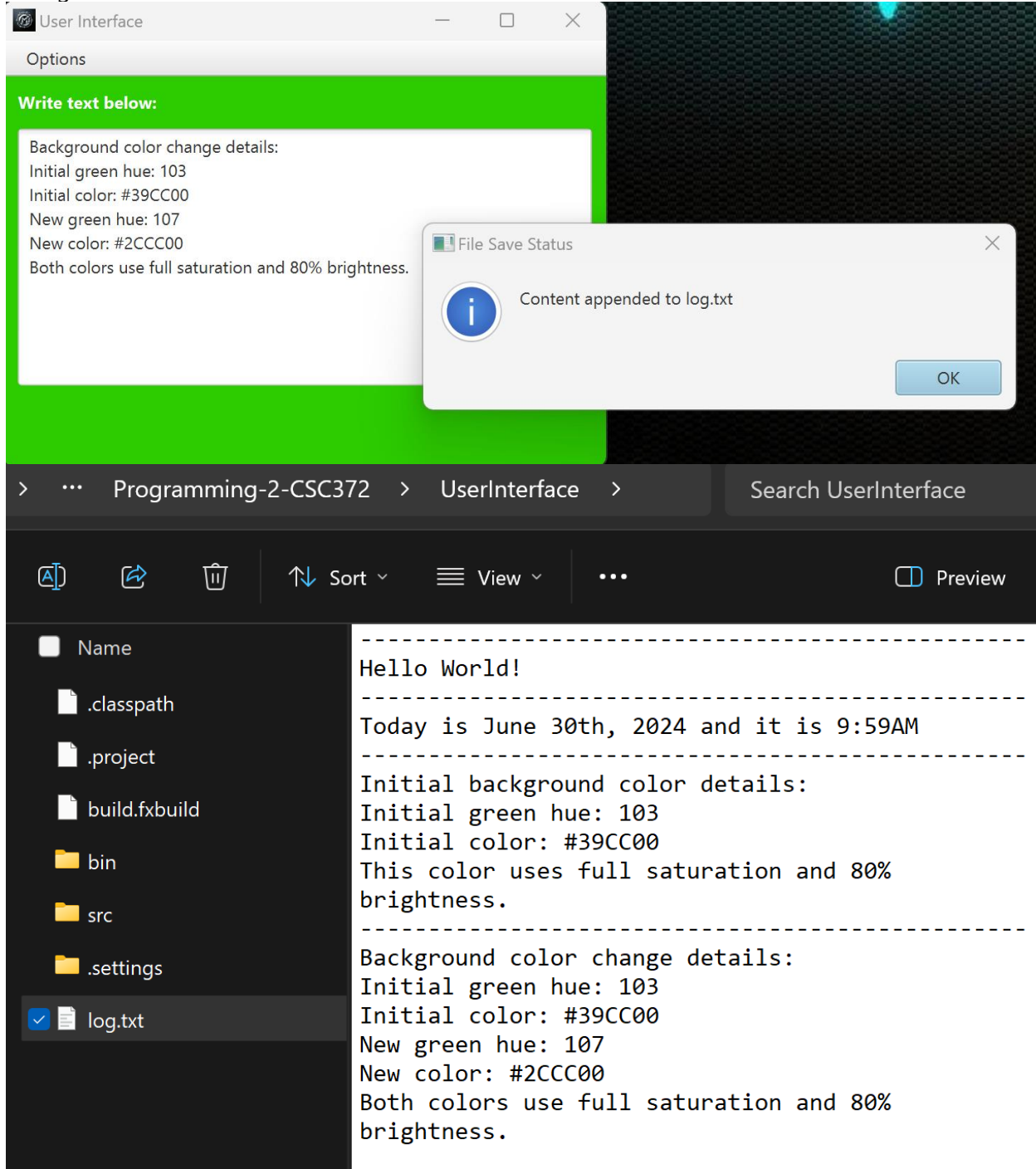


Note: After choosing the option “Save to File,” the date and time was appended to the file.

Figure 8*Background Collor Initial Green Hue*

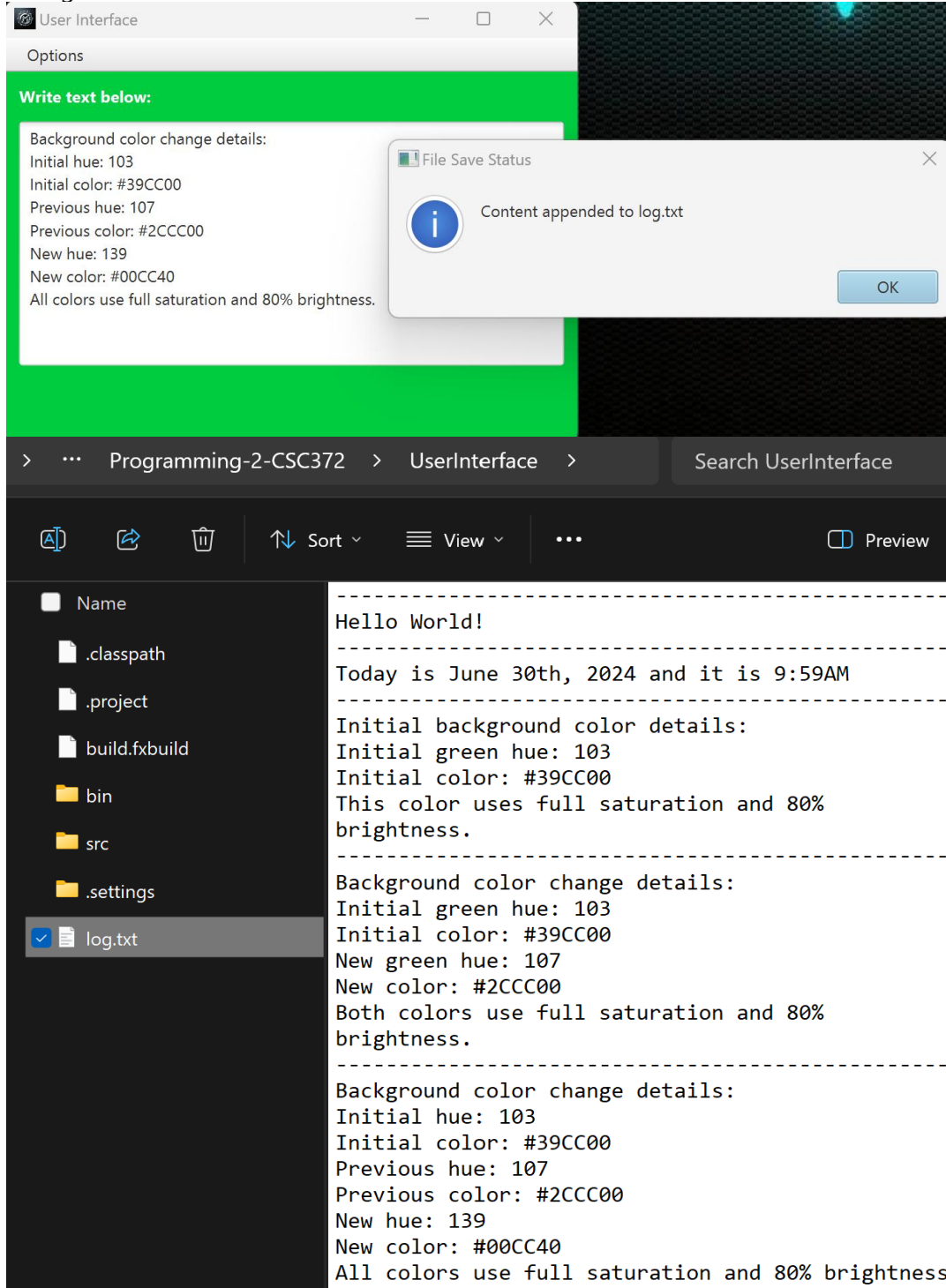
Note: After choosing the option “Save to File,” the change to background color was appended to the file.

More next page

Figure 9*Background Collor New Green Hue*

Note: After choosing the option “Save to File,” the change to background color was appended to the file.

Figure 10
Background Collor More New Green Hue



Note: After choosing the option “Save to File,” the change to background color was appended to the file.

As shown in Figure 1 through Figure 10 the functionalities tests run without any issues, displaying the correct outputs as expected.